

ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

SEMESTER PROJECT SPRING 2025

MASTER IN COMPUTATIONAL SCIENCE AND ENGINEERING

Projected Langevin Dynamics for Optimal Transport

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Contents

1	Introduction	1
2	Sinkhorn Algorithm	1
2.1	Basic Optimal Transport theory	1
2.2	Entropic OT and Sinkhorn Algorithm	2
2.3	Numerical application: Gaussian distribution	5
3	Projected Langevin Dynamics	6
3.1	Model formulation	6
3.2	Convergence results	8
3.3	Example : Centered gaussian marginals	9
4	Conditional expectation estimation	11
4.1	Nadaraya-Watson estimator	11
4.2	Polynomial regression	12
4.3	k-NN Monte-Carlo	13
4.4	Rate of convergence and Bias-Variance decomposition	15
5	Numerical Experiments	16
5.1	Comparison of conditional expectation estimation methods	16
5.2	Error convergence and Runtimes	19
5.3	LSI Bound	20
6	Conclusion	21
	Appendix A Algorithms	23
	Appendix B Figures	24

1 Introduction

Optimal transport is a relevant topic across different theoretical and applied fields, including machine-learning, optimization, and gradient flows. While the classical linear programme algorithms has prohibitive complexity of $\mathcal{O}(N^3)$ (with N as the histogram dimension), its entropic relaxation has gained popularity by reducing the complexity to near quadratic, depending on the regularization factor. The resulting Sinkhorn algorithm is the method of choice for majority of computational optimal transport problems. Despite the success of Sinkhorn algorithm, the computational cost and memory requirement for practical estimation of optimal transport maps and Wasserstein distances remain significant.

Recently, Conforti et al. [6] have proposed a system of Langevin dynamics whose stationary law coincides with the solution of the entropic optimal transport, for continuous marginal distributions. While their work remains purely theoretical, the novel projected-Langevin dynamics can pave the way for innovative numerical schemes addressing the optimal transport problem.

This study addresses computational investigation of projected-Langevin dynamics aiming to provide insight into its numerical difficulties and potential advantages in comparison to popular Sinkhorn algorithm.

We start by providing an overview on practical Sinkhorn variants in Section 2: Sinkhorn-Knopp, Greenkhorn, Projected-Sinkhorn with rounding and ε -scaling with log-domain stabilisation. We also clarify the accuracy-complexity trade-offs that guide their use.

Section 3 formulates the projected Langevin dynamics analyzing how they preserve the marginal constraints and converge exponentially fast under specific assumptions. An explicit Euler-Maruyama (EM) discretisation is derived. A central technical obstacle is the repeated evaluation of conditional expectations that drive the drift.

Section 4 proposes three estimators to solve this problem: Nadaraya-Watson kernels, ordinary-least-squares polynomial regression and k-nearest-neighbour Monte Carlo, and analyses their theoretical complexity, bias-variance decomposition and dimensional scaling properties.

Section 5 embeds these estimators into the EM scheme and benchmarks the resulting solver against ε -scaling and Greenkhorn on Gaussian marginals. This setting is practical as it admits closed-form Bures-Wasserstein and entropic costs and therefore enables a rigorous error assessment.

The numerical results demonstrate that, especially for low ε , the projected-Langevin scheme augmented with either k-nearest-neighbour or ordinary-least-squares estimators yields markedly lower transport error than both ε -scaling and Greenkhorn, while offering superior per-step complexity: OLS advances in $\mathcal{O}(N)$ operations and kNN in $\mathcal{O}(N \log N)$, whereas each Sinkhorn iterate remains quadratic in the sample size. Within the Gaussian benchmark OLS is both the most accurate and the fastest, yet its polynomial ansatz is ill-suited to non-Gaussian or highly structured couplings, where the non-parametric kNN regression exhibits greater robustness.

2 Sinkhorn Algorithm

2.1 Basic Optimal Transport theory

We first start with a probabilistic formulation of the Optimal-Transport (OT) problem, originally formalized by Monge in 1781. Two measures (μ, ν) , supported on two spaces (X, Y) , are linked through a measurable map $T : X \rightarrow Y$ that fulfills (see [7])

$$\min_T \int_X c(x, T(x)) d\mu(x) \quad \text{subject to } T_{\#}\mu = \nu. \quad (1)$$

The constraint $T_{\#}\mu = \nu$ means that T *pushes forward* the measure μ to ν , using the pushforward operator as defined in [17]. $c(x, y)$ denotes the cost and throughout this report we will focus on the Euclidean distance $c(x, y) = \|x - y\|^2$ which is strictly convex. This formulation can be ill-posed as some times there is no T satisfying $T_{\#}\mu = \nu$, for example in the case where μ is a Dirac measure but ν is not. This was coped by Kantorovich who relaxed the problem and allowed for the mass to be split in the modified formulation seeking

$$\inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathbb{R}^{2d}} c(x, y) \pi(dx, dy). \quad (2)$$

Here, $\Pi(\mu, \nu)$ denotes the set all probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ whose first marginal is μ and the second marginal is ν . A crucial result (see [2]) is that there always exist a minimizer for this problem when the cost function $c(x, y)$ is lower semi-continuous and $\Pi(\mu, \nu)$ is a tight collection of measures (we will use Euclidean d -spaces $X = Y = \mathbb{R}^d$ for which tightness is guaranteed). Expression 2 corresponds to the Wasserstein distance of order p via

$$W_p(\mu, \nu) = \left(\inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathbb{R}^{2d}} \|x - y\|^p \pi(dx, dy) \right)^{1/p}. \quad (3)$$

By leveraging the marginal constraints from the Kantorovich formulation, we can transform the problem into a dual formulation, where $\varphi(x)$ and $\psi(y)$ are the respective Schrödinger potentials over X and Y which act as Lagrange multipliers leading to (see [17])

$$\sup_{\varphi, \psi} \left\{ \int_X \varphi(x) d\mu(x) + \int_Y \psi(y) d\nu(y) \right\} \quad (4)$$

subject to

$$\varphi(x) + \psi(y) \leq c(x, y) \quad \text{for all } x \in X, y \in Y$$

Network simplex and interior point methods have been derived and applied to the primal formulation 2, but have proven to be computationally costly. Typically the cost of computing Wasserstein distances scales as $\mathcal{O}(d^3 \log d)$ when comparing two histograms of dimension d , or two point clouds each containing d points, in a general metric space (see [16]). This can become cheaper in the particular case where the metric probability space of interest can be embedded into $\mathbb{R}^n$ with $n \leq 4$ (see [15]). The main question of our interest is: How can we make the OT problem computationally efficient?

Two crucial ideas have been employed: entropy regularization and the use of Schrodinger potentials. These two constitute the foundation of the Sinkhorn algorithm.

2.2 Entropic OT and Sinkhorn Algorithm

The first idea is to regularize the transport with a Kullback-Leibler divergence and seek the coupling

$$\pi_\epsilon^* = \arg \min_{\pi \in \Pi(\mu, \nu)} \int c(x, y) \pi(dx, dy) + \epsilon H(\pi \| \mu \otimes \nu), \quad (5)$$

where ϵ is the regularization strength and $H(\pi \| \mu \otimes \nu) = \int_{\mathcal{X} \times \mathcal{Y}} \log \left(\frac{d\pi}{d(\mu \otimes \nu)}(x, y) \right) d\pi(x, y)$ is the KL divergence. The entropy term penalizes deviations of π_ϵ from the independent (trivial) coupling $\mu \otimes \nu$, encouraging smoother transport plans.

We will now focus on giving a heuristic derivation of the Sinkhorn method to better understand why it has become the popular algorithm for optimal transport computations. The algorithm assumes a discrete setting which is why we consider two discrete probability measures

$$\alpha = \sum_{i=1}^n a_i \delta_{x_i}, \quad \beta = \sum_{j=1}^m b_j \delta_{y_j},$$

where $\{x_i\}_{i=1}^n \subset \mathcal{X}$, $\{y_j\}_{j=1}^m \subset \mathcal{Y}$, and $a \in \mathbb{R}_+^n$, $b \in \mathbb{R}_+^m$ are probability vectors such that $\sum_i a_i = \sum_j b_j = 1$. The coupling matrix $P \in \mathbb{R}_+^{n \times m}$, which is equivalent to the transport plan π in the continuous setting, has each entry $P_{i,j}$ representing the amount of mass transported from x_i to y_j . Admissible couplings are solely those that preserve the marginals, forming the following transport polytope

$$\mathcal{U}(a, b) := \{P \in \mathbb{R}_+^{n \times m} \mid P \mathbf{1}_m = a, P^\top \mathbf{1}_n = b\}. \quad (6)$$

Now the minimal (Wasserstein) distances for the OT problem $\mathcal{L}_C(a, b)$ and the entropic OT problem $\mathcal{L}_C^\epsilon(a, b)$ are the following:

$$\mathcal{L}_C(a, b) := \min_{P \in \mathcal{U}(a, b)} \langle C, P \rangle = \sum_{i=1}^n \sum_{j=1}^m C_{i,j} P_{i,j}, \quad \mathcal{L}_C^\epsilon(a, b) := \min_{P \in \mathcal{U}(a, b)} \langle C, P \rangle - \epsilon H(P) \quad (7)$$

where the cost matrix $C \in \mathbb{R}^{n \times m}$ is defined entrywise by $C_{i,j} := c(x_i, y_j) = \|x_i - y_j\|^2$. An important result is that the unique solution P_ε of the entropically regularized problem 7 converges, as $\varepsilon \rightarrow 0$, to the optimal solution of the Kantorovich problem that maximizes entropy [17](Prop. 4.1):

$$P_\varepsilon \xrightarrow{\varepsilon \rightarrow 0} \arg \min \{ -H(P) : P \in \mathcal{U}(a, b), \langle P, C \rangle = \mathcal{L}_C(a, b) \} \Leftrightarrow s\mathcal{L}_C^\varepsilon(a, b) \xrightarrow{\varepsilon \rightarrow 0} \mathcal{L}_C(a, b) \quad (8)$$

Equation 5 can be refactored as a projection problem [17](Prop. 4.2), i.e.

$$\min_{\pi \in \mathcal{U}(\alpha, \beta)} H(\pi \| \mathcal{K}) \quad \mathcal{K} \text{ is the Gibbs distribution } d\mathcal{K}(x, y) := e^{-c(x, y)/\varepsilon} d\alpha(x) d\beta(y), \quad (9)$$

Using this formulation in the discretized case and introducing the two dual variables $\varphi \in \mathbb{R}^n$ and $\psi \in \mathbb{R}^m$ corresponding to the marginal constraints, the Lagrangian associated with the entropically regularized optimal transport problem 7 becomes

$$\mathcal{E}(P, \varphi, \psi) = \langle P, C \rangle - \varepsilon H(P) - \langle \varphi, P \mathbf{1}_m - a \rangle - \langle \psi, P^\top \mathbf{1}_n - b \rangle. \quad (10)$$

The Gibbs kernel $\mathcal{K}$, plays a very important role for the convergence of the Sinkhorn algorithm: from Theorem 2.1 of [10] we have that a necessary and sufficient condition that the Sinkhorn-Knopp algorithm converges is that $\mathcal{K}$ has total support. Or equivalently, every row and column contains at least one positive entry and the matrix is "fully indecomposable", meaning there is no way to permute its rows and columns as to obtain a block-upper-triangular form with a zero block off the diagonal (for more details see [10]). The first-order optimality conditions yield

$$\frac{\partial \mathcal{E}(P, \varphi, \psi)}{\partial P_{i,j}} = C_{i,j} + \varepsilon \log(P_{i,j}) - \varphi_i - \psi_j = 0 \quad (11)$$

which implies that the optimal transport plan P satisfies

$$P_{i,j} = e^{\varphi_i/\varepsilon} e^{-C_{i,j}/\varepsilon} e^{\psi_j/\varepsilon} = u_i \mathcal{K}_{ij} v_j. \quad (12)$$

This derivation from [17](Prop 4.3) establishes the deep connection between the Schrödinger potentials φ and ψ and the structure of the optimal transport plan P . However, since these potentials are unknown, the practical focus shifts to computing the scaling vectors u and v . As $P = \text{diag}(u) \mathcal{K} \text{diag}(v)$, the variables (u, v) must satisfy the following nonlinear equations, which enforce the marginal constraints inherent to $\mathcal{U}(a, b)$, i.e.

$$\text{diag}(u) \mathcal{K} \text{diag}(v) \mathbf{1}_m = a \quad \text{and} \quad \text{diag}(v) \mathcal{K}^\top \text{diag}(u) \mathbf{1}_n = b \quad (13)$$

which reduce to

$$u \odot (Kv) = a, \quad \text{and} \quad v \odot (K^\top u) = b,$$

where $\odot$ denotes the element-wise product. The way Sinkhorn's algorithm handles these equations is by solving them iteratively

$$u^{(\ell+1)} := \frac{a}{Kv^{(\ell)}} \quad v^{(\ell+1)} := \frac{b}{K^\top u^{(\ell+1)}} \quad (14)$$

where division is understood element-wise. The algorithm is initialized with an arbitrary positive vector (see Algorithm 1 for the outline).

Algorithm 1: Sinkhorn-Knopp Algorithm (SK)

```

1 Input:  $a, b, C, \lambda$ 
   Input:  $\mathbf{u}^{(0)} = \mathbf{1}, \quad K = \exp(-\frac{C}{\lambda})$ 
2 for  $i = 1, \dots, n_{it}$  do
3    $\mathbf{v}^{(i)} = b \odot (K^\top \mathbf{u}^{(i-1)})^{-1}$  ; // Update right scaling
4    $\mathbf{u}^{(i)} = a \odot (K \mathbf{v}^{(i)})^{-1}$  ; // Update left scaling
5 return  $T = \text{diag}(\mathbf{u}^{(n_{it})}) K \text{diag}(\mathbf{v}^{(n_{it})})$ 
```

This algorithm is implemented using the `sinkhorn.bregman.sinkhorn_knopp` function from the Python Optimal Transport (POT) library [8].

¹Source: <http://remi.flamary.com/cours/otml/sink.png>

Convergence analysis

For the convergence and time complexity of the Sinkhorn algorithm, *Altschuler et al.* [1] showed that by setting $\epsilon = 4 \log(n)/\tau$, $\mathcal{O}(\|C\|_\infty^3 \log(n) \tau^{-3})$ Sinkhorn iterations, with the additional rounding step to compute a valid coupling $\hat{P} \in U(a, b)$, are enough to ensure $\langle \hat{P}, C \rangle \leq \mathcal{L}_C(a, b) + \tau$. More precisely, Sinkhorn computes a τ -approximate solution of the unregularized OT problem in $\mathcal{O}(n^2 \log(n) \tau^{-3})$ operations [17]. The rounding scheme is applied just after running the regular Sinkhorn-Knopp algorithm 1 and consists of

$$u' = u \odot \min\left(\frac{a}{u \odot (Kv)}, \mathbf{1}\right) \quad v' = v \odot \min\left(\frac{b}{v \odot (K^T u')}, \mathbf{1}\right) \quad (15)$$

$$\Delta_a = a - u' \odot (Kv') \quad \Delta_b = b - v' \odot (K^T u') \quad (16)$$

$$\hat{P} = \text{diag}(u') K \text{diag}(v') + \frac{\Delta_a \Delta_b^T}{\|\Delta_a\|_1} \quad (17)$$

where u' and v' define the "clipped" scalings, and Δ_a, Δ_b are the marginal residuals that tell us how far $\text{diag}(u') K \text{diag}(v')$ is from having the exact marginals a and b . So we have that $\hat{P} \in U(a, b)$ and that: $\|\hat{P} - \text{diag}(u') K \text{diag}(v')\|_1 \leq \|a - u \odot (Kv)\|_1 + \|b - v \odot (K^T u)\|_1$. This is implemented in Python following pseudo-algorithm 2 in [1].

Greenkhorn

In the same paper, the authors introduce an even faster algorithm: Greenkhorn (see Alg. 4 [1]). This is a greedy coordinate-descent variant of Sinkhorn-Knopp where instead of alternately normalizing all rows then all columns, we pick at each step the single row or column whose current marginal violation is largest and updates only that coordinate. If we note the current plan as $P^{(t)} = \text{diag}(u^{(t)}) K \text{diag}(v^{(t)})$ and define the row and column residuals as:

$$\begin{cases} R_i = r_i - \sum_j P_{ij}^{(t)} = r_i - [u^{(t)} \odot (K v^{(t)})]_i \implies i^* = \arg \max_i |R_i| \\ C_j = c_j - \sum_i P_{ij}^{(t)} = c_j - [v^{(t)} \odot (K^T u^{(t)})]_j \implies j^* = \arg \max_j |C_j| \end{cases} \quad (18)$$

where the update now becomes

$$\begin{cases} |R_{i^*}| \geq |C_{j^*}| \implies u_{i^*}^{(t+1)} = u_{i^*}^{(t)} \frac{r_{i^*}}{\sum_j K_{i^*j} v_j^{(t)}}, & v^{(t+1)} = v^{(t)} \\ |R_{i^*}| < |C_{j^*}| \implies v_{j^*}^{(t+1)} = v_{j^*}^{(t)} \frac{c_{j^*}}{\sum_i K_{ij^*} u_i^{(t)}}, & u^{(t+1)} = u^{(t)} \end{cases} \quad (19)$$

Despite updating only one coordinate at a time, Greenkhorn still converges geometrically to the unique entropic-OT plan and can be combined with the same final rounding step (Algorithm 2) to recover an exact coupling in $\mathcal{U}(r, c)$ with the same tight cost error bounds. Greenkhorn matches the best known Sinkhorn rate with a complexity $\tilde{\mathcal{O}}(n^2 \epsilon^{-2})$ (Sinkhorn-Knopp: $\tilde{\mathcal{O}}(n^2 \epsilon^{-3})$) while often performing fewer total operations in practice [12]. This is implemented using the `sinkhorn.bregman.greenkhorn` function from the POT library [8].

ϵ -Scaling and Log-Domain Computations

The idea for ϵ -scaling is to solve a sequence of less-regularized problems, using each solution to warm-start the next (see [5]). We first define an outer-loop:

$$\epsilon^{(k)} = (\epsilon_0 - \epsilon) e^{-k} + \epsilon, \quad k = 0, 1, 2, \dots, K \quad (20)$$

where $\epsilon_0 \gg \epsilon$ is a large initial value. At each stage k , we now solve the entropic OT problem with regularization $\epsilon^{(k)}$. Empirically, only a few inner iterations of Sinkhorn are needed per stage and the optimal coupling of one stage is used as a warm-start for the next one. The second strategy that is used to avoid numerical collapse when ϵ is small or C has large entries, is to compute α and β as the log-analogs of u and v ($u_i = \exp(\frac{\alpha_i}{\epsilon})$ and $v_j = \exp(\frac{\beta_j}{\epsilon})$) (see [18]). To do this, for each row index

$i = 1, \dots, n$ and $j = 1, \dots, m$ at stage k we have

$$\alpha_i \leftarrow \varepsilon^{(k)} \left[\ln a_i - \ln \sum_j \exp\left(\frac{\alpha_i + \beta_j - C_{ij}}{\varepsilon^{(k)}}\right) \right], \quad (21)$$

$$\beta_j \leftarrow \varepsilon^{(k)} \left[\ln b_j - \ln \sum_i \exp\left(\frac{\alpha_i + \beta_j - C_{ij}}{\varepsilon^{(k)}}\right) \right], \quad (22)$$

and if the log-scaling vectors still drift outside the floating-point range we can recenter it by subtracting the maximum of the entire vector: if $\max_{1 \leq i \leq n} |\alpha_i| > \tau$ then $\alpha \leftarrow \alpha - \max_{1 \leq i \leq n} \alpha_i$ and if $\max_{1 \leq j \leq m} |\beta_j| > \tau$ then $\beta \leftarrow \beta - \max_{1 \leq j \leq m} \beta_j$, where τ is defined by the user. Then we use the final (α, β) from stage k as the initialization for stage $k + 1$. This is numerically implemented using `ot.bregman.sinkhorn.epsilon.scaling` from the POT library [8].

2.3 Numerical application: Gaussian distribution

For the following numerical experiment, we consider two 1-dimensional gaussian distributions $\mu = \mathcal{N}(20, 5)$, $\nu = \mathcal{N}(60, 10)$. The exact Wasserstein distance is computed using the Bures-Wasserstein analytical formula (see [11]) for Gaussians introduced in Eq. 44. We also compute the entropic Optimal transport distance $\mathbf{OT}_2^\varepsilon(\mu, \nu)$ using the result given from Theorem 1:

Theorem 1 (Entropic 2-Wasserstein for Gaussians, Corollary 1(a) in [13]). *Let $\mu_0 = \mathcal{N}(m_0, K_0)$ and $\mu_1 = \mathcal{N}(m_1, K_1)$ be two Gaussian measures on $\mathbb{R}^n$, with means $m_0, m_1 \in \mathbb{R}^n$ and covariance matrices $K_0, K_1 \in \mathbb{R}^{n \times n}$. For any entropic regularization parameter $\varepsilon > 0$, define*

$$M^\varepsilon = I_n + \sqrt{I_n + \frac{16}{\varepsilon^2} K_0 K_1} \in \mathbb{R}^{n \times n}.$$

Then the entropic-regularized 2-Wasserstein cost

$$OT_{d^2}^\varepsilon(\mu_0, \mu_1) = \inf_{P \in \Pi(\mu_0, \mu_1)} \langle P, \|x - y\|^2 \rangle + \varepsilon H(P)$$

admits the closed-form

$$\begin{aligned} OT_{d^2}^\varepsilon(\mu_0, \mu_1) &= \|m_0 - m_1\|^2 + \text{Tr}(K_0) + \text{Tr}(K_1) \\ &\quad - \frac{\varepsilon}{2} \left(\text{Tr}(M^\varepsilon) - \ln \det(M^\varepsilon) + n \ln 2 - 2n \right). \end{aligned}$$

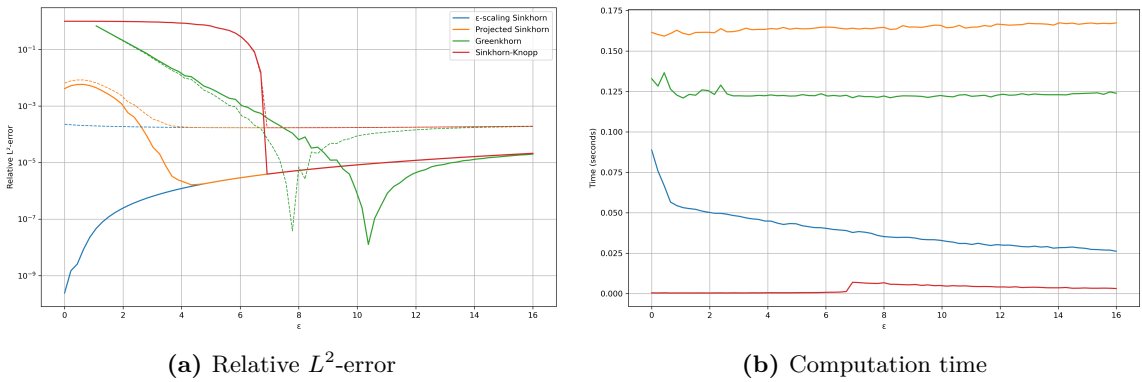


Figure 1: (a) Relative L^2 -error of the four Sinkhorn algorithms as a function of ε . Solid curves plot $\|\langle P, C \rangle - \mathcal{W}_{\text{BW}}\|^2 / \|\mathcal{W}_{\text{BW}}\|^2$, where $\mathcal{W}_{\text{BW}}$ is the Bures-Wasserstein distance and dashed curves plot $\|\langle P, C \rangle - OT^\varepsilon\|^2 / \|OT^\varepsilon\|^2$. (b) Computation time (in seconds) versus ε .

Greenkhorn has very large error at tiny ε as the Gibbs kernel K is ill-conditioned, it struggles to enforce the marginals. Ill conditioning of the Gibbs kernel here refers not to its spectral condition number, but to its element-wise dynamic range $\frac{\max_{i,j} K_{ij}}{\min_{i,j} K_{ij}} = \exp\left(\frac{C_{\max}}{\varepsilon}\right)$. This is illustrated in Table 1, where we report the element-wise measure $\frac{C_{\max}}{\varepsilon}$ instead of $\exp(C_{\max}/\varepsilon)$ to avoid `inf` values

when exponentiating. When ε increases to moderate values (≈ 2 – 10), the problem becomes better conditioned and Greenhorn converges more reliably with its error falling sharply. Beyond $\varepsilon \approx 10$, the entropic bias grows, so the relative error rises again. Classical Sinkhorn–Knopp likewise fails at small ε for the same reason as Greenhorn and improves with a very sharp fall-off in its error once ε increases. Then the error slowly increases again because of the entropic bias. While the Projected-Sinkhorn algorithm uses the exact same multiplicative updates as Sinkhorn–Knopp, the “rounding” of the coupling $\hat{P}$ before evaluating the cost, removes the marginal violations which introduce a bias when not evaluating on a feasible plan. The ε -scaling maintains a rapidly decreasing error even for $\varepsilon = 0.001$ where all other methods fail to do so. It is the only method robust at all ε delivering high accuracy for small regularization and competitive speed as it leverages the gradual smoothing and warm-starts (see Eqs.20, 21). Moreover, as ε becomes large, all four methods’ error curves collapse onto one another, demonstrating that they are consistent.

Table 1: Element-wise condition measure $C_{\max}/\varepsilon$ at selected ε .

ε	0.001	0.217	1.082	5.190	10.163	16.000
$\frac{C_{\max}}{\varepsilon}$	9.801×10^6	4.512×10^4	9.058×10^3	1.888×10^3	9.644×10^2	6.126×10^2

3 Projected Langevin Dynamics

3.1 Model formulation

In recent study [6], it is shown that the probability law induced by the following Langevin dynamics converges to the optimal coupling π of 5 when the marginals have log-density potentials $\mu(dx) = e^{-U(x)} dx$ and $\nu(dy) = e^{-V(y)} dy$ (under some mild assumptions). The so-called projected Langevin dynamics follows

$$\begin{cases} dX_t = \left(\mathbb{E}[\nabla_x c(X_t, Y_t) | X_t] - \nabla_x c(X_t, Y_t) - \varepsilon \nabla U(X_t) \right) dt + \sqrt{2\varepsilon} dB_t, \\ dY_t = \left(\mathbb{E}[\nabla_y c(X_t, Y_t) | Y_t] - \nabla_y c(X_t, Y_t) - \varepsilon \nabla V(Y_t) \right) dt + \sqrt{2\varepsilon} dW_t. \end{cases} \quad (23)$$

where B_t and W_t are independent d -dimensional Brownian motions. We first start by giving a heuristic explanation for this system in the case where we have quadratic $c(x, y) = |x - y|^2$. The scheme then becomes

$$\begin{cases} dX_t = (Y_t - \mathbb{E}[Y_t | X_t] - \varepsilon \nabla U(X_t)) dt + \sqrt{2\varepsilon} dW_t, \\ dY_t = (X_t - \mathbb{E}[X_t | Y_t] - \varepsilon \nabla V(Y_t)) dt + \sqrt{2\varepsilon} dB_t. \end{cases} \quad (24)$$

We recall that the Ornstein-Uhlenbeck process has the following form

$$d\bar{X}_t = -\varepsilon \nabla U(\bar{X}_t) dt + \sqrt{2\varepsilon} dW_t \quad (25)$$

where $U : \mathbb{R}^d \rightarrow \mathbb{R}$ is a quadratic potential, so that $\nabla U(x) = Ax$ for some positive-definite matrix A . Furthermore note that the invariant (stationary) probability distribution μ is given by

$$\mu(dx) = \frac{e^{-U(x)}}{\int_{\mathbb{R}^d} e^{-U(z)} dz} dx. \quad (26)$$

The generator of this invariant measure for a smooth function $\varphi(x)$ is written as:

$$\mathcal{L}_\mu \varphi(x) = -\varepsilon \nabla U(x) \cdot \nabla \varphi(x) + \varepsilon \Delta \varphi(x).$$

from which we obtain the following important property:

$$\frac{d}{dt} \mathbb{E}_\mu[\varphi(\bar{X}_t)] = \mathbb{E}_\mu[\mathcal{L}_\mu \varphi] = 0 \quad (27)$$

Now imagine running two independent copies obeying the same process, one for X_t and one for Y_t . Their joint law is frozen at the independent coupling $\mu \otimes \nu$. Hence, inside the OT polytope $\Pi(\mu, \nu) = \{P \in \mathcal{P}(\mathbb{R}^{2d}) : P^X = \mu, P^Y = \nu\}$ this system would explore only a single point which is useless if we are trying to approximate the optimal coupling $\pi \neq \mu \otimes \nu$. The cross-terms $Y_t - \mathbb{E}[Y_t | X_t]$ and $X_t - \mathbb{E}[X_t | Y_t]$ inject correlations between the two coordinates allowing them to "wander" through the entire polytope $\Pi(\mu, \nu)$ instead of being stuck at $\mu \otimes \nu$. Now the question is whether the marginals are still conserved. We consider the projected system (Eq.24) by adding the cross term $Y_t - \mathbb{E}[Y_t | X_t]$, apply Itô's formula and then take the expectation:

$$\frac{d}{dt} \mathbb{E}[\varphi(X_t)] = \mathbb{E}[(Y_t - \mathbb{E}[Y_t | X_t]) \cdot \nabla \varphi(X_t)] + \mathbb{E}[\mathcal{L}_\mu \varphi(X_t)]$$

If we pose $Z_t = Y_t - \mathbb{E}[Y_t | X_t]$ we obtain

$$\mathbb{E}[Z_t \cdot \nabla \varphi(X_t)] = \mathbb{E}[\mathbb{E}[Z_t \cdot \nabla \varphi(X_t) | X_t]] = \mathbb{E}[\nabla \varphi(X_t) \cdot \mathbb{E}[Z_t | X_t]] = 0 \quad (28)$$

as $\mathbb{E}[Z_t | X_t] = \mathbb{E}[Y_t - \mathbb{E}[Y_t | X_t] | X_t] = 0$. Hence, the marginals are conserved and the system is well-posed. This is a very heuristic outline but is the crux of the whole construction. For a more detailed and rigorous derivation and proof see Conforti et al. [6] (section 2: for an analysis from a gradient flow point of view and the geometry of $\Pi(\mu, \nu)$; section 4: for a detailed of the cross-terms and more specifically their convergence to the Schrödinger potentials). We now recall the central theorem of [6] that turns this informal construction into a rigorously well-posed system.

Theorem 2 (Theorem 1.1 from [6]). *Suppose the functions (c, U, V) are twice continuously differentiable, such that $\mu(dx) = e^{-U(x)} dx$ and $\nu(dy) = e^{-V(y)} dy$ are probability measures. Assume the following:*

- **Sub-Gaussian marginals:** *There exists $\delta > 0$ such that $\int_{\mathbb{R}^d} e^{\delta|x|^2} (e^{-U(x)} + e^{-V(x)}) dx < \infty$*
- **Bounded Hessians:** *$\nabla^2 c$ is bounded, and $\nabla^2 U, \nabla^2 V$ are bounded from below in the semidefinite order.*
- **Integrability:** *$|\nabla U| \in L^p(\mu), |\nabla V| \in L^p(\nu)$ for some $p > 2(d+1)$, and $U \in L^1(\mu), V \in L^1(\nu)$.*

Then, for any $P_0 \in \Pi(\mu, \nu)$, the following hold:

1. *There exists a unique (in law) weak solution of the coupled SDE system 23, starting from $(X_0, Y_0) \sim P_0$.*
2. *The law P_t of (X_t, Y_t) satisfies $P_t \in \Pi(\mu, \nu)$ for each $t \geq 0$.*
3. *As $t \rightarrow \infty$, we have $P_t \rightarrow \pi$ in 2-Wasserstein distance, and $\mathcal{H}(P_t | \pi) \rightarrow 0$, where π is the entropic OT plan.*
4. *As $t \rightarrow \infty$, the optimal transport gradients are recovered:*

$$\int_{\mathbb{R}^d} \|\mathbb{E}[\nabla_x c(X_t, Y_t) | X_t = x] - \nabla \varphi(x)\|^2 \mu(dx) \rightarrow 0,$$

$$\int_{\mathbb{R}^d} \|\mathbb{E}[\nabla_y c(X_t, Y_t) | Y_t = y] - \nabla \psi(y)\|^2 \nu(dy) \rightarrow 0.$$

These requirements will be respected in all numerical simulations throughout this report as we will focus on the quadratic cost $c(x, y) = |x - y|^2$ and choose Gaussian marginals. Having established existence and uniqueness, the coupled SDE system in (23) must still be solved numerically. The *Euler–Maruyama* discretisation is the canonical choice: it is easy to implement, has strong order 1/2, and thanks to its explicit form, it extends straightforwardly to high-dimensional particle systems.

Algorithm 2: Euler–Maruyama Projected Langevin Dynamics

Input: $X^{(0)}, Y^{(0)}, \varepsilon, T, N$ // Initial particles, regularization, horizon, steps
1 $\Delta t \leftarrow T/N$ // Time step size
2 $\sigma \leftarrow \sqrt{2\varepsilon \Delta t}$ // Brownian noise scale
3 **for** $n = 0$ **to** $N - 1$ **do**
4 $M_{YX} \leftarrow \text{COND_MEAN_METHOD}(X^{(n)}, Y^{(n)})$ // $\mathbb{E}[Y | X]$
5 $M_{XY} \leftarrow \text{COND_MEAN_METHOD}(Y^{(n)}, X^{(n)})$ // $\mathbb{E}[X | Y]$
6 $G_U \leftarrow \text{GRAD_U}(X^{(n)})$ // Compute ∇U
7 $G_V \leftarrow \text{GRAD_V}(Y^{(n)})$ // Compute ∇V
8 $\Delta B, \Delta W \sim \mathcal{N}(0, \sigma^2 I)$ // Brownian increments
9 $X^{(n+1)} \leftarrow X^{(n)} + (Y^{(n)} - M_{YX} - \varepsilon G_U)\Delta t + \Delta B$
10 $Y^{(n+1)} \leftarrow Y^{(n)} + (X^{(n)} - M_{XY} - \varepsilon G_V)\Delta t + \Delta W$
11 **return** $\{X^{(n)}, Y^{(n)}\}_{n=0}^N$

The number of samples is denoted by N_s , the dimension of the marginals by d , and N is the number of timesteps made by the model. In terms of memory, storing the trajectory arrays is the most expensive $X_{\text{traj}}, Y_{\text{traj}} \in \mathbb{R}^{N_s \times d \times (N+1)}$, as storing the gradients G_U, G_V , conditional expectations M_{YX}, M_{XY} and Brownian increments takes $N_s \times d$ memory. Therefore in the case where we want to store all the time-steps in the trajectory, the memory scales as $\mathcal{O}(N_s \times d \times N)$, and if not (the calculations are made in-place) it scales as $\mathcal{O}(N_s \times d)$. In terms of operations, we have two calls at each timestep for noise generation costing and the vector updates costing $\mathcal{O}(N_s \times d)$ each. These are necessary for the scheme to work so we know that the lower bound in terms of execution time will scale as $\mathcal{O}(N_s \times d \times N)$. The calculation of the gradients will also be fairly simple and will not surpass $N_s \times d$. The question now is how to make the estimation of the conditional expectations as efficient as possible while maintaining good accuracy. This remark is crucial, as even small errors may accumulate across iterations and lead to instabilities or run-away behavior in the scheme.

3.2 Convergence results

In this section we address the exponential convergence properties of 24 (always in the case of quadratic cost). In [6] (section 4), the authors prove the following energy dissipation identity

$$\frac{d}{dt} \mathcal{J}(P_t) = \varepsilon \frac{d}{dt} H(P_t | \pi) = -\varepsilon^2 \bar{I}(P_t | \pi) \quad (29)$$

where $J(P)$ denotes the energy functional $\mathcal{J}(P) = \int c(x, y) P(dx, dy) + \varepsilon H(P | \mu \otimes \nu)$ along the flow $(P_t)_{t \geq 0}$, and for $P \in \Pi(\mu, \nu)$, $R = \log(dP/d\pi)$ we also define

$$\bar{I}(P | \pi) = \mathbb{E}_P \left[\|\nabla_x R - \mathbb{E}[\nabla_x R | X]\|^2 + \|\nabla_y R - \mathbb{E}[\nabla_y R | Y]\|^2 \right] \quad (30)$$

To prove an exponential rate of convergence, they bound $\bar{I}(\cdot | \pi)$ from below in terms of $H(\cdot | \pi)$, which leads them to consider the following projected logarithmic Sobolev inequality

$$H(P | \pi) \leq \kappa \bar{I}(P | \pi), \quad \forall P \in \Pi(\mu, \nu). \quad (31)$$

Recall that we say that π satisfies the logarithmic Sobolev inequality with constant $\kappa > 0$, or LSI(κ), if

$$H(P | \pi) \leq \kappa^{-1} I(P | \pi), \quad \forall P \in \mathcal{P}(\mathbb{R}^{2d}) \quad (\text{LSI}(\kappa))$$

Here, $I(P | \pi) := \|\nabla \log \frac{dP}{d\pi}\|_{L^2(P)}^2$ is the usual relative Fisher information, which is defined to be ∞ if the gradient does not exist in $L^2(P)$. As such, the projected LSI can be interpreted as a lower bound on the deficit of the standard LSI, quantifying how much the inequality can be tightened by leveraging the coupling structure of P . For a more detailed analysis, we refer the reader to Section 4 of [6]. The following theorem gives a sufficient condition for the projected LSI 31 (Theorem 1.2 of [6])

Theorem 3 (Theorem 1.2 from [6]). *Assume that there exist constants $\kappa^{X|Y}$ and $\kappa^{Y|X}$ such that:*

1. *The conditional measure $\pi^x(dy) = \pi(dx, dy)/\mu(dx)$ satisfies $\text{LSI}(\kappa^{X|Y})$, uniformly in $y \in \mathbb{R}^d$,*

2. The conditional measure $\pi^y(dx) = \pi(dx, dy)/\nu(dy)$ satisfies $\text{LSI}(\kappa^{Y|X})$, uniformly in $x \in \mathbb{R}^d$.

If, moreover,

$$\left(\kappa^{X|Y} \kappa^{Y|X}\right)^{1/2} \varepsilon^{-1} > 2, \quad (32)$$

then the projected LSI (31) holds with

$$r = \varepsilon \left(\kappa^{X|Y} + \kappa^{Y|X} \right) \left(1 - \frac{4}{\varepsilon^2 \kappa^{X|Y} \kappa^{Y|X}} \right). \quad (33)$$

As a consequence, for any initialization $P \in \Pi(\mu, \nu)$ of the SDE (24), we have exponential convergence in entropy:

$$H(P_t | \pi) \leq e^{-rt} H(P_0 | \pi), \quad \text{for all } t \geq 0. \quad (34)$$

Moreover, the gradients of the Schrödinger potentials converge exponentially as well:

$$\int_{\mathbb{R}^d} \|\mathbb{E}[X_t - Y_t | X_t = x] - \nabla \varphi(x)\|^2 \mu(dx) \leq \frac{4}{\kappa^{Y|X}} e^{-rt} H(P_0 | \pi), \quad (35)$$

$$\int_{\mathbb{R}^d} \|\mathbb{E}[Y_t - X_t | Y_t = y] - \nabla \psi(y)\|^2 \nu(dy) \leq \frac{4}{\kappa^{X|Y}} e^{-rt} H(P_0 | \pi). \quad (36)$$

We also present the corollary of Thm. 3 specifically for strongly log-concave marginals, which encompasses the gaussian case analyzed in Section 5.3.

Corollary 4 (Corollary 1.3 of [6]). Assume the potentials $U, V: \mathbb{R}^d \rightarrow \mathbb{R}$ are twice continuously differentiable and that there exist

$$\alpha_U, \alpha_V \in (0, \infty), \quad \beta_U, \beta_V \in (0, \infty]$$

such that

$$\alpha_U I \preceq \nabla^2 U(x) \preceq \beta_U I, \quad \alpha_V I \preceq \nabla^2 V(y) \preceq \beta_V I, \quad \text{for all } x, y \in \mathbb{R}^d.$$

If

$$\varepsilon > \frac{\alpha_U^{-1} \alpha_V^{-1} - \beta_U^{-1} \beta_V^{-1}}{(\alpha_U^{-1} + \beta_U^{-1})^{1/2} (\alpha_V^{-1} + \beta_V^{-1})^{1/2}} =: \varepsilon_c, \quad (37)$$

then the projected logarithmic Sobolev inequality (31) holds with the rate r from (33) and with²

$$\kappa^{X|Y} = \sqrt{4\alpha_U/(\varepsilon^2 \beta_V) + \alpha_V^2} + \alpha_U, \quad \kappa^{Y|X} = \sqrt{4\alpha_V/(\varepsilon^2 \beta_U) + \alpha_U^2} + \alpha_V. \quad (38)$$

Symmetric case. If $\alpha_U = \alpha_V = \alpha$ and $\beta_U = \beta_V = \beta$, then (37) simplifies to

$$\varepsilon > \alpha^{-1} - \beta^{-1}.$$

Under (37), the exponential L^2 -convergence estimates of equations 35 and 36 also hold.

3.3 Example : Centered gaussian marginals

If we let μ and ν be centered Gaussian measures with covariance matrices Σ_μ and Σ_ν , which are given $d \times d$ positive definite matrices, the SDE (23) corresponding to the quadratic cost function becomes

$$\begin{aligned} dX_t &= (Y_t - \mathbb{E}[Y_t | X_t] - \varepsilon \Sigma_\mu^{-1} X_t) dt + \sqrt{2\varepsilon} dB_t, \\ dY_t &= (X_t - \mathbb{E}[X_t | Y_t] - \varepsilon \Sigma_\nu^{-1} Y_t) dt + \sqrt{2\varepsilon} dW_t. \end{aligned} \quad (39)$$

We also have the following properties for gaussian measures

$$\mathbb{E}[Y_t | X_t] = \Sigma_t^\top \Sigma_\mu^{-1} X_t, \quad \mathbb{E}[X_t | Y_t] = \Sigma_t \Sigma_\nu^{-1} Y_t \quad (40)$$

for $\Sigma_t = \mathbb{E}[X_t Y_t^\top]$. By Substituting into (39), the SDE becomes:

²We adopt the convention $\beta^{-1} = 0$ when $\beta = \infty$.

$$\begin{aligned} dX_t &= (Y_t - (\Sigma_t^\top + \varepsilon I) \Sigma_\mu^{-1} X_t) dt + \sqrt{2\varepsilon} dB_t, \\ dY_t &= (X_t - (\Sigma_t + \varepsilon I) \Sigma_\nu^{-1} Y_t) dt + \sqrt{2\varepsilon} dW_t. \end{aligned} \quad (41)$$

We now consider the specific case where the two marginal distributions are one-dimensional $\mu = \mathcal{N}(0, \sigma_0^2)$, $\nu = \mathcal{N}(0, \sigma_1^2)$ the cross-covariance becomes $\Sigma_t = \mathbb{E}[X_t Y_t]$. Substituting this into the projected Langevin system, the coupled SDEs become:

$$\begin{aligned} dX_t &= (Y_t - \sigma_0^{-2} (\mathbb{E}[X_t \cdot Y_t] + \varepsilon) X_t) dt + \sqrt{2\varepsilon} dB_t \\ dY_t &= (X_t - \sigma_1^{-2} (\mathbb{E}[X_t \cdot Y_t] + \varepsilon) Y_t) dt + \sqrt{2\varepsilon} dW_t \end{aligned} \quad (42)$$

This 1-dimensional Gaussian setting serves as a tractable example to study the numerical behavior of the scheme and analyze its convergence properties. At this stage, we do not yet estimate the conditional expectations $\mathbb{E}[Y_t | X_t]$ and $\mathbb{E}[X_t | Y_t]$ from data; instead, we exploit their closed-form expressions available in the Gaussian case. From point (2) of Theorem 2, we know that the law P_t of the process (X_t, Y_t) remains in the coupling set $\Pi(\mu, \nu)$ for all $t \geq 0$. This means that although the dynamics evolve in time, the marginal distributions of X_t and Y_t remain fixed as μ and ν . This means that we can use the following empirical estimator for the p-Wasserstein distance (in 1-dimensional setting) [17] to track the distance between X_t and Y_t and estimate the convergence to the optimal transport coupling:

$$\widehat{\mathcal{W}}_p^\varepsilon(\mu, \nu)_t = \left(\frac{1}{n} \sum_{i=1}^n |X_t^{(i)} - Y_t^{(i)}|^p \right)^{1/p} \quad (43)$$

where $\{X_t^{(i)}\}_{i=1}^n$ and $\{Y_t^{(i)}\}_{i=1}^n$ are the order statistics (naturally coupled so no need to implement the statistic) of the independent samples drawn from the empirical distributions $\hat{\mu}$ and $\hat{\nu}$ at time t .

We will use this estimator in the 1D setting to approximate the Wasserstein distance between X_t and Y_t at each time step, and compare it with the exact theoretical value known from the Gaussian case. The squared 2-Wasserstein distance $W_2^2(\mu, \nu)$ between two Gaussian distributions $\mu = \mathcal{N}(m_0, \Sigma_0)$ and $\nu = \mathcal{N}(m_1, \Sigma_1)$, is called the Bures-Wasserstein distance and admits the following closed-form expression [11]:

$$W_2^2(p_0, p_1) = \|m_0 - m_1\|^2 + \mathcal{B}^2(\Sigma_0, \Sigma_1) \quad (44)$$

where the squared Bures distance between covariance matrices is given by:

$$\mathcal{B}^2(\Sigma_0, \Sigma_1) = \text{tr} \left(\Sigma_0 + \Sigma_1 - 2 \left(\Sigma_0^{1/2} \Sigma_1 \Sigma_0^{1/2} \right)^{1/2} \right) \quad (45)$$

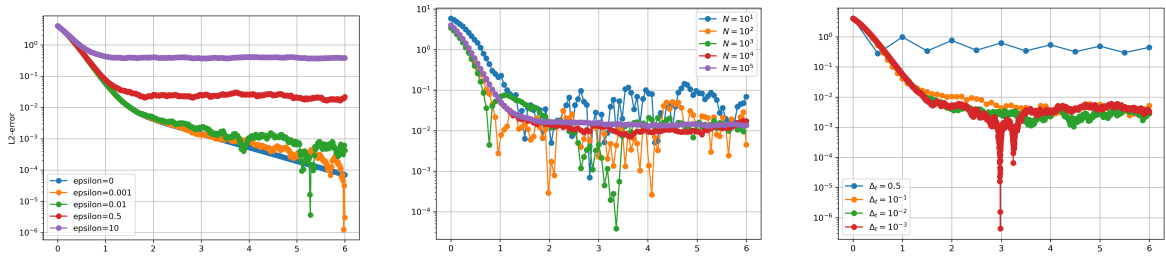


Figure 2: Pointwise relative L^2 -error for marginals $\mu = \mathcal{N}(0, 5)$ and $\nu = \mathcal{N}(0, 10)$, namely $|\widehat{W}_2^2(X_t, Y_t) - W_2^2(\mu, \nu)|^2 / W_2^4(\mu, \nu)$, under separate variations of the regularization parameter ε , the sample size N , and the time-step Δt .

As ε increases from 0 (blue) to 10 (purple), the error plateau shifts upward. For $\varepsilon = 0$, the estimator converges monotonically to zero and continues to decrease indefinitely (see left-most Figure 2 for a zoom on the $\varepsilon = 0$ case). The other epsilon values are shown in the left-most Figure 2 to converge to the error between their respective entropic Wasserstein distances. Increasing the number of samples N reduces Monte-Carlo fluctuations and lowers the error floor. With $N = 10$ (blue), the relative error remains around 10^{-1} – 10^0 , whereas for $N = 10^5$ (purple), it decays smoothly to 10^{-2} – 10^{-3} . We also observe that $N = 10^5$ and $N = 10^4$ produce almost identical error trajectories,

with $N = 10^5$ exhibiting only slightly reduced fluctuations compared to $N = 10^4$. We observe that for $\Delta t \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ the convergence curves are nearly indistinguishable, indicating that Δt has little influence on the error decay; the only exception is the largest time-step $\Delta t = 0.5$, which is substantial relative to the horizon $T = 6$ and produces noticeable oscillations.

4 Conditional expectation estimation

4.1 Nadaraya-Watson estimator

We start by recalling that the conditional expectation can be written as a regression function $m(x)$:

$$m(x) = \mathbb{E}[Y \mid X = x] = \int y f_{Y|X=x}(y) dy = \frac{\int y f(x, y) dy}{f_X(x)}. \quad (46)$$

We can deduce that given samples $\{(X_i, Y_i)\}_{i=1}^n$, the regression function can be computed from the joint density f and the marginal f_X by replacing these densities by their kernel density estimators (KDE) [14]. We consider that the joint distribution takes the form

$$\hat{f}(x, y; \mathbf{h}) = \frac{1}{n} \sum_{i=1}^n K_{h_1}(x - X_i) K_{h_2}(y - Y_i), \quad (47)$$

where $K_h(\cdot)$ denotes a kernel function with bandwidth h and $\mathbf{h} = (h_1, h_2)$, and considering the same bandwidth h_1 the KDE of f_X becomes

$$\hat{f}_X(x; h_1) = \frac{1}{n} \sum_{i=1}^n K_{h_1}(x - X_i). \quad (48)$$

the estimator of m can be defined by replacing the KDEs ³ 47 and 48 in 46:

$$\hat{m}(x) = \frac{\int y \hat{f}(x, y; h) dy}{\hat{f}_X(x; h_1)} = \frac{\int y \frac{1}{n} \sum_{i=1}^n K_{h_1}(x - X_i) K_{h_2}(y - Y_i) dy}{\frac{1}{n} \sum_{i=1}^n K_{h_1}(x - X_i)} \quad (49)$$

$$= \frac{\frac{1}{n} \sum_{i=1}^n K_{h_1}(x - X_i) \int y K_{h_2}(y - Y_i) dy}{\frac{1}{n} \sum_{i=1}^n K_{h_1}(x - X_i)} \quad (50)$$

$$= \sum_{i=1}^n \frac{K_{h_1}(x - X_i)}{\sum_{i=1}^n K_{h_1}(x - X_i)} Y_i \quad (51)$$

In the d -dimensional case, $\{([X_{1i}, \dots, X_{di}], [Y_{1i}, \dots, Y_{di}])\}_{i=1}^n$ this formula becomes:

$$\hat{m}(\mathbf{x}) = \sum_{i=1}^n \frac{K_{\mathbf{H}}(\mathbf{x} - \mathbf{X}_i)}{\sum_{i=1}^n K_{\mathbf{H}}(\mathbf{x} - \mathbf{X}_i)} \mathbf{Y}_i \quad (52)$$

where $\mathbf{x} \in \mathbb{R}^d$, $\mathbf{H} \in \mathbb{R}^{d \times d}$ and $K : \mathbb{R}^d \rightarrow \mathbb{R}$ such that: $K_{\mathbf{H}}(\mathbf{u}) = |\mathbf{H}|^{-1/2} K(\mathbf{H}^{-1/2} \mathbf{u})$. $\mathbf{H}$ is the symmetric positive-definite bandwidth matrix and can be computed via the eigendecomposition $\mathbf{H}^{-1/2} = \mathbf{P} \mathbf{\Lambda}^{-1/2} \mathbf{P}^\top$ if $\mathbf{H} = \mathbf{P} \mathbf{\Lambda} \mathbf{P}^\top$. The kernel evaluation for a single point costs $\mathcal{O}(d)$ and if we have N query points at N sample locations the cost becomes $\mathcal{O}(N^2 d)$. We present the algorithmic implementation of the Nadaraya-Watson estimator in Alg.3, where we elect to leave out the computation of the weight when the query is equal to the sample.

³<https://bookdown.org/egarpor/PM-UC3M/npreg-kre.html>

Algorithm 3: Nadaraya–Watson Estimator (Leave-One-Out)

Input: $\mathbf{X} = [X_1, \dots, X_n]$, $\mathbf{Y} = [Y_1, \dots, Y_n]$, $h > 0$, $K(\cdot)$
Output: $\hat{\mathbf{m}} = [\hat{m}_1, \dots, \hat{m}_n]$

```
1 for  $j = 1$  to  $n$  do
2   Num  $\leftarrow 0$ ;
3   Den  $\leftarrow 0$ ;
4   for  $i = 1$  to  $n$  do
5     if  $i = j$  then
6       continue ;           // Leave-one-out: skip using  $X_j$  to predict itself
7     end
8      $w \leftarrow K\left(\frac{X_j - X_i}{h}\right)$ ;
9     Num  $\leftarrow$  Num +  $w \cdot Y_i$ ;
10    Den  $\leftarrow$  Den +  $w$ ;
11  end
12  if Den = 0 then
13     $\hat{m}_j \leftarrow 1$ ;
14  else
15     $\hat{m}_j \leftarrow$  Num / Den;
16  end
17 end
18 return  $\hat{\mathbf{m}}$ 
```

4.2 Polynomial regression

It is known that the conditional expectation $\mathbb{E}[Y | X]$ is the minimizer of the mean squared distance over all L^2 -functions $m : \mathbb{R}^d \rightarrow \mathbb{R}$ (see [3], [4]): $\mathbb{E}[(Y - m(X))^2]$. In our application, we suppose that the joint distribution of X and Y is not known precisely and the minimization problem above cannot be solved exactly. In this case the idea is to approximate the true function $f(X)$ via least squares regression, consisting in minimizing the empirical mean squared error:

$$\frac{1}{N_s} \sum_{n=1}^{N_s} (Y^{(n)} - m(X^{(n)}))^2 \quad (53)$$

based on samples $(X^{(n)}, Y^{(n)})$ of the pair (X, Y) , over a suitable family $\mathcal{S}$ of Borel functions $m : \mathbb{R}^d \rightarrow \mathbb{R}$. We will consider the second-order polynomial regression, meaning we regress on $1, X_1, \dots, X_d$ and all second-order terms $X_i X_j$, for $1 \leq i \leq j \leq d$. The corresponding function family $\mathcal{S}$ is the linear span of $\{1, x_1, \dots, x_d, x_i x_j \mid 1 \leq i \leq j \leq d\}$. The feature matrix A has N_s rows and $P = 1 + \frac{3d}{2} + \frac{d^2}{2}$ columns and the regression function takes the form:

$$\hat{m}(x) = \hat{\beta}_0 + \sum_{i=1}^d \hat{\beta}_i x_i + \sum_{1 \leq i \leq j \leq d} \hat{\beta}_{ij} x_i x_j \quad (54)$$

The parameters $\hat{\beta} \in \mathbb{R}^P$ are obtained by solving the least squares problem:

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^P} \frac{1}{M} \sum_{m=1}^M (Y^{(m)} - m_{\beta}(X^{(m)}))^2 \quad (55)$$

The normal equation is used to compute $\hat{\beta}$ numerically, by computing the singular value decomposition of A :

$$A^{\top} A \hat{\beta} = A^{\top} Y \Leftrightarrow \hat{\beta} = V \Sigma^{-1} U^{\top} Y \quad (56)$$

When A is rank-deficient, the normal equation admits infinitely many solutions. The QR algorithm will choose one such solution, but it does not control the Euclidean norm $\|\beta\|_2$, so β_{QR} may be large or poorly behaved. By contrast, the SVD pseudoinverse $\beta_{\text{SVD}} = A^+ y$ where $A^+ = V \Sigma^{-1} U^{\top}$ yields the unique solution of minimal ℓ_2 -norm. This acts as an implicit regulariser in an equivalent way as ridge regression. As a consequence, there is an additional cost of $\mathcal{O}(MP^2 + P^3)$ floating-point operations,

compared to QR which only has $\mathcal{O}(MP^2)$. We used the `sklearn.linear_model.LinearRegression` function.

Algorithm 4: Second-Order Polynomial Regression via SVD

Input: $A \in \mathbb{R}^{M \times P}$, $Y \in \mathbb{R}^M$
Output: $\hat{\beta} \in \mathbb{R}^P$, $\hat{m}(x)$

- 1 $[U, \Sigma, V^\top] \leftarrow \text{SVD}(A)$; // $A = U\Sigma V^\top$
- 2 $A^+ \leftarrow V\Sigma^{-1}U^\top$
- 3 $\hat{\beta} \leftarrow A^+Y$
- 4 $\hat{m}(x) \leftarrow A\hat{\beta}$
- 5 **return** $\hat{\beta}$, $\hat{m}(x)$

4.3 k-NN Monte-Carlo

Let $(X_i, Y_i)_{i=1}^n$ be i.i.d. observations with $X_i \in \mathbb{R}^d$, $Y_i \in \mathbb{R}$ and $m(x) = \mathbb{E}[Y \mid X = x]$. For a query point $x \in \mathbb{R}^d$, we pose that $\mathcal{N}_k(x) = \{i_1(x), \dots, i_k(x)\} \subset \{1, \dots, n\}$ are the indices of the k closest X_i 's to x in Euclidean distance. The k -nearest-neighbour regression estimator is the vector average [20]

$$\hat{m}_{n,k}(x) = \frac{1}{k} \sum_{j \in \mathcal{N}_k(x)} Y_j \quad (57)$$

If the regression surface $m(\cdot)$ is smooth, then points whose predictors lie in a small ball $B(x, r) = \{u \in \mathbb{R}^d : \|u - x\| \leq r\}$ have very similar conditional means. Therefore, averaging the corresponding Y -values should yield a good approximation to $m(x)$. The k -nearest-neighbour (k-NN) method sets the radius adaptively by defining $r_{n,k}(x) := \|x - X_{(k)}(x)\|$ where $X_{(k)}(x)$ denotes the k -th closest point to x among the sample $\{X_i\}_{i=1}^n$. The k-NN algorithm consists of first computing the Euclidean distance $\|x - X_i\|$ for all $i = 1 \dots n$, finding the indices of the k smallest distances and finally outputting the mean. For a single query this consists of computing n distances in d dimensions which is $\mathcal{O}(nd)$ complexity and the sorting takes $\mathcal{O}(n \log n)$. For n queries (all the X_i) the total time becomes $\mathcal{O}(n^2d)$ which does not offer a real advantage compared to N-W estimator. We can improve on this by making use of a tree data structure called the KD-tree (k-dimensional tree). The tree is constructed by recursively splitting the dataset along the median of alternating coordinate axes yielding a binary tree where each node corresponds to hyperrectangle (see Alg. 5 from [19]). At each node, we choose the coordinate $j \in \{1, \dots, d\}$ where the data has the largest spread

$$j = \arg \max_{\ell \in [d]} \left(\max_i X_i^{(\ell)} - \min_i X_i^{(\ell)} \right) .$$

Then we find the median point X_m of the dataset X along dimension j , which defines a splitting hyperplane $H = \{X \in \mathbb{R}^d \mid X^{(j)} = X_m^{(j)}\}$. We obtain in the left subtree all points with $X^{(j)} < X_m^{(j)}$ and in the right one all points with $X^{(j)} > X_m^{(j)}$. Each node stores a single point and recursively stores its left/right partitions (see Fig. 3). Median selection can be done in linear time, so the total construction satisfies (Complexity recursion can be solved using ⁴):

$$T(n) = \Theta(n) + 2 \cdot T\left(\left\lceil \frac{n}{2} \right\rceil\right) \Rightarrow T(n) = \Theta(n \log n).$$

⁴Master's Theorem

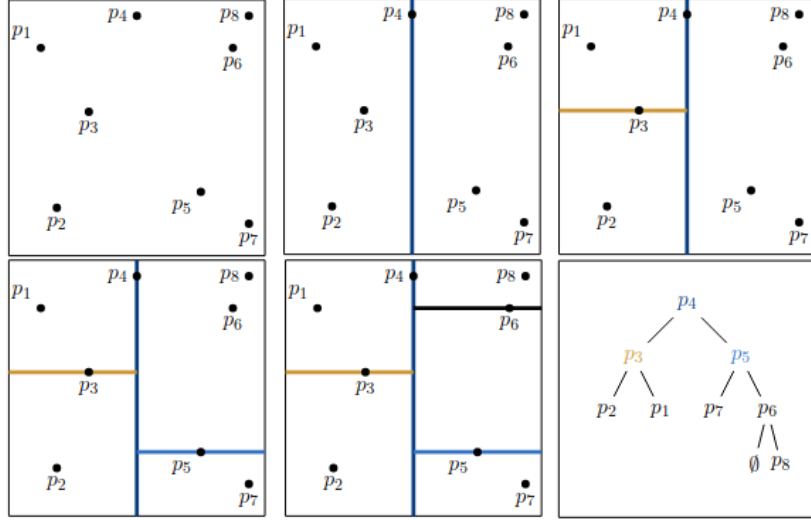


Figure 3: Recursively building a k-d tree on eight points. The hyperplanes are shown in the first five subfigures, while the full binary tree representation is shown in the last subfigure (bottom-right). Source: adapted from [19]

For the querying procedure, we first start by comparing the query point X to the current node's splitting coordinate and going left if X is less than the split value or right otherwise. This is repeated until reaching a leaf node. At each visited node, check whether the current point is within the k closest to X and if so, update the list of candidates. To do this we keep a list L of the best k nearest neighbors found so far. If the list becomes longer than k , the farthest point from X is removed. The real key of this algorithm is checking the splitting hyperplane: after searching one side, we check whether the other side of the hyperplane could contain closer points and if the distance from X to the hyperplane is smaller than the farthest distance in L , we recurse into the other subtree (for a more detailed analysis see [19]). Once the algorithm is terminated we return final list L of the k closest neighbors to the query point X .

Algorithm 5: BUILD KD-TREE [19]

Input: Data points $(X_1, \dots, X_n) \subset \mathbb{R}^d$
Output: KD-tree over the input points

```

1 if  $n \leq 1$  then
2   | return node containing  $(X_1, \dots, X_n)$ 
3 end
4  $j \leftarrow \arg \max_{\ell \in [d]} (\max_i X_i^{(\ell)} - \min_i X_i^{(\ell)})$ ;           // coordinate with max spread
5  $X_m \leftarrow$  median of  $(X_1, \dots, X_n)$  along dimension  $j$ 
6  $L \leftarrow \{X_i \mid X_i^{(j)} < X_m^{(j)}\}$ ;                             // left subset
7  $R \leftarrow \{X_i \mid X_i^{(j)} > X_m^{(j)}\}$ ;                             // right subset
8  $H \leftarrow \{X \in \mathbb{R}^d \mid X^{(j)} = X_m^{(j)}\}$ ;                     // splitting hyperplane
9  $T_L \leftarrow$  BUILD-KD-TREE( $L$ )
10  $T_R \leftarrow$  BUILD-KD-TREE( $R$ )
11 return node containing  $X_m$  and  $H$ , with  $T_L, T_R$  as children

```

The full recursive procedure NN-KD-TREE-QUERY for querying is detailed in Appendix 7 for completeness.

Algorithm 6: Regression estimation using k-NN with KD-tree

Input: Query points $X_1, \dots, X_n \in \mathbb{R}^d$

Sample points $Y_1, \dots, Y_N \in \mathbb{R}^d$

Responses $Z_1, \dots, Z_N \in \mathbb{R}$

Number of neighbors k

Output: Estimated conditional means $\hat{m}(X_j) = \mathbb{E}[Z \mid X_j]$ for $j = 1, \dots, n$

1 $T \leftarrow \text{BUILD-KD-TREE}(Y_1, \dots, Y_N)$

2 **for** $j = 1$ **to** n **do**

3 $L_j \leftarrow \text{NN-KD-TREE-QUERY}(X_j, T, \varepsilon, k, \emptyset)$; // find k -NN of X_j in Y

4 $\hat{m}(X_j) \leftarrow \frac{1}{k} \sum_{i \in L_j} Z_i$; // average the responses

5 **end**

6 **return** $\hat{m}(X_1), \dots, \hat{m}(X_n)$

4.4 Rate of convergence and Bias-Variance decomposition

Theorem 5 (Theorem 6.1 from [9]). *Let $k = k(n) \rightarrow \infty$ and $k/n \rightarrow 0$ as $n \rightarrow \infty$. Then the k -nearest neighbor regression estimator satisfies:*

$$\int (\hat{m}_{n,k}(x) - m(x))^2 dP_X(x) \xrightarrow{P} 0$$

That is, the k -NN estimator is $L_2(P_X)$ -consistent for every distribution of (X, Y) with $\mathbb{E}[Y^2] < \infty$.

Theorem 5 shows the power of the k-NN estimator: no assumptions are made on $m(x)$, no density assumption on X and it holds for every distribution of (X, Y) as long as $\mathbb{E}[Y^2] < \infty$. The same exact theorem is valid for the Nadaraya-Watson estimator with the additions that: $K : \mathbb{R}^d \rightarrow \mathbb{R}$ is a bounded, non-negative, Lipschitz, and integrable function satisfying $\int K(u) du > 0$ and the bandwidth h_n satisfies $h_n \rightarrow 0$ and $nh_n^d \rightarrow \infty$ as $n \rightarrow \infty$. If we suppose that the regression function m is Lipschitz continuous (or in a Sobolev class of smoothness $\beta = 1$), and the marginal density f_X is bounded, K is bounded, symmetric, Lipschitz kernel with finite second moment, then the mean-squared error satisfies (see [20]):

$$\mathbb{E}[(\hat{m}_n(x) - m(x))^2] = \mathcal{O}(n^{-2/(2+d)}) \quad (58)$$

which is the minimax-optimal rate for nonparametric regression on Lipschitz functions in dimension d and where for N-W estimator and k-NN, we have respectively $h \asymp n^{-1/(2+d)}$ and $k \asymp n^{2/(2+d)}$. The above error rate $n^{-2/(2+d)}$ exhibits a very poor dependence on the dimension d because if we consider a small $\varepsilon > 0$ and we want n to verify that $n^{-2/(2+d)} \leq \varepsilon$, we obtain that $n \geq \varepsilon^{-(2+d)/2}$. This is the curse of dimensionality: as the dimension d increases, we require exponentially more samples n in order to achieve an error bound of ε . We now examine the mean-squared prediction error and L_2 -consistency of the OLS estimator (see 10.2 of [9]).

Theorem 6 (Mean-squared prediction error of OLS and consistency). *Let $\{(X_i, Y_i)\}_{i=1}^n \subset \mathbb{R}^p \times \mathbb{R}$ be i.i.d. observations from $Y_i = X_i^\top \beta^* + \varepsilon_i$ with $i = 1, \dots, n$ where we suppose that:*

1. $\mathbb{E}[\varepsilon_i \mid X_i] = 0$ and $\text{Var}(\varepsilon_i) = \sigma^2 < \infty$;
2. the design has full second-moment matrix $\Sigma_X := \mathbb{E}[X_i X_i^\top] \succ 0$.

Denoting by $\hat{\beta} = (X^\top X)^{-1} X^\top Y$ (see Eq.56) the ordinary least-squares estimator and set $\hat{m}_n(x) = x^\top \hat{\beta}$. Then the error is bounded as:

$$\mathbb{E}[(X^\top \hat{\beta} - X^\top \beta^*)^2] \leq \sigma^2 p/n$$

Consequently $\|\hat{m}_n - m\|_{L_2(P_X)}^2 = O(pn^{-1}) \Rightarrow \lim_{n \rightarrow \infty} \|\hat{m}_n - m\|_{L_2(P_X)}^2 = 0$.

Unlike the non-parametric estimators k -NN and Nadaraya-Watson that make no structural assumption on m (*universally consistent*), ordinary least squares relies on a correctly specified linear model. In Theorem 6 we assumed that the regression function satisfies $m(x) = x^\top \beta^*$. If the true regression has the more general form $m(x) = \varphi(x)$ with $\varphi : \mathbb{R}^p \rightarrow \mathbb{R}$ not contained in the linear

span of the design columns then the error bound and consistency guarantee from Th.6 no longer hold. Now we examine the bias-variance decomposition for these three methods. The mean-squared prediction error splits as: $\mathbb{E}[(\hat{m}(x) - m(x))^2] = \text{Var}[\hat{m}(x)] + (\mathbb{E}[\hat{m}(x)] - m(x))^2$ where the first one is the variance and the second is the bias squared. If the model is correctly specified then the bias term; $\text{Bias}[\hat{m}(x)] = x^\top \beta^* - m(x)$ becomes zero and we have the asymptotic rate for the variance from Th 6 of $\mathcal{O}(p/n)$ (see Table 2).

Estimator	Variance	Squared bias
OLS	$\mathcal{O}(p/n)$	0 (correct model)
Nadaraya–Watson ⁴	$\mathcal{O}(1/(n h^d))$	$\mathcal{O}(h^{2\beta})$
k -NN ⁴	$\mathcal{O}(1/k)$	$\mathcal{O}((k/n)^{2\beta/d})$

Table 2: Variance–bias orders when the regression function is β -Hölder smooth (here $\beta = 1$) and the design density is bounded in dimension d .

We can also see from Table 2 that the rate of variance for NW depends on the scaled bandwidth h^{-d} while for kNN it only depends on the number of neighbours k . Heuristically, no matter what the dimension is, the ball $B(x, r_k(x))$ always contains k observations. Thus the variability of k-NN is only caused by the k neighbouring points independently of the dimension. On the other hand, for NW a certain value of h will cover different numbers of observations in different dimensions. This shows the advantage of k-NN as k needs tuning only regarding the number of samples and not the dimension, and it also performs better generally in high-dimensions.

5 Numerical Experiments

5.1 Comparison of conditional expectation estimation methods

We now embed these conditional-expectation estimators into our projected Langevin simulation via an explicit Euler–Maruyama discretization (see Section 3). We will focus on Gaussian and exponential families, since for these distributions most key quantities such as the Bures–Wasserstein distance admit exact, closed-form expressions. We first start by analyzing a bivariate normal distribution:

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}\left(\underbrace{\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}}_{\mu}, \underbrace{\begin{pmatrix} \sigma_1^2 & \rho \sigma_1 \sigma_2 \\ \rho \sigma_1 \sigma_2 & \sigma_2^2 \end{pmatrix}}_{\Sigma}\right),$$

where $\sigma_1, \sigma_2 > 0$ and $\rho \in (-1, 1)$. Standard theory tells us that:

$$m(x) = \mathbb{E}[Y | X = x] = \mu_2 + \rho \frac{\sigma_2}{\sigma_1} (x - \mu_1) \quad (59)$$

In fact, in the Gaussian–to–Gaussian case the Brenier transport map coincides exactly with the regression function $m(x)$ which is what we use in Figs. 4 (black line).

OLS, being a polynomial (linear) regressor, matches the linear Gaussian map almost exactly here (though it can fail on non-polynomial targets, e.g. Gaussian→Exponential, see Figure 11 top row). The nonparametric methods (middle and bottom) also recover the map with high fidelity, but exhibit small deviations near the boundaries.

In the left figure of 5 we can see that kNN achieves a good error rate although it increases slowly towards the end. But from the rightmost figure which is a snapshot at $t = 20$, we can see that a few points at the boundaries of the coupling have totally strayed away (the scale is at $1e6$). We have to zoom out from the last figure at $t = 20$ of Fig. 4 where the majority of points do respect the coupling but the behavior of the outliers is not captured at the scale of the figure. This decoupling can also be seen in the rightmost figure where the scale is at $1e6$. As the dynamics evolve the paired points begin to stretch away toward infinity, causing the error to grow without bound; this runaway divergence is much more apparent in the generated animations. It is important to note that in reality what is important here is not the time but the number of iterations and therefore the timestep. Therefore

⁴See MIT lecture notes: https://www.mit.edu/~9.520/fall118/slides/Class18_local.pdf

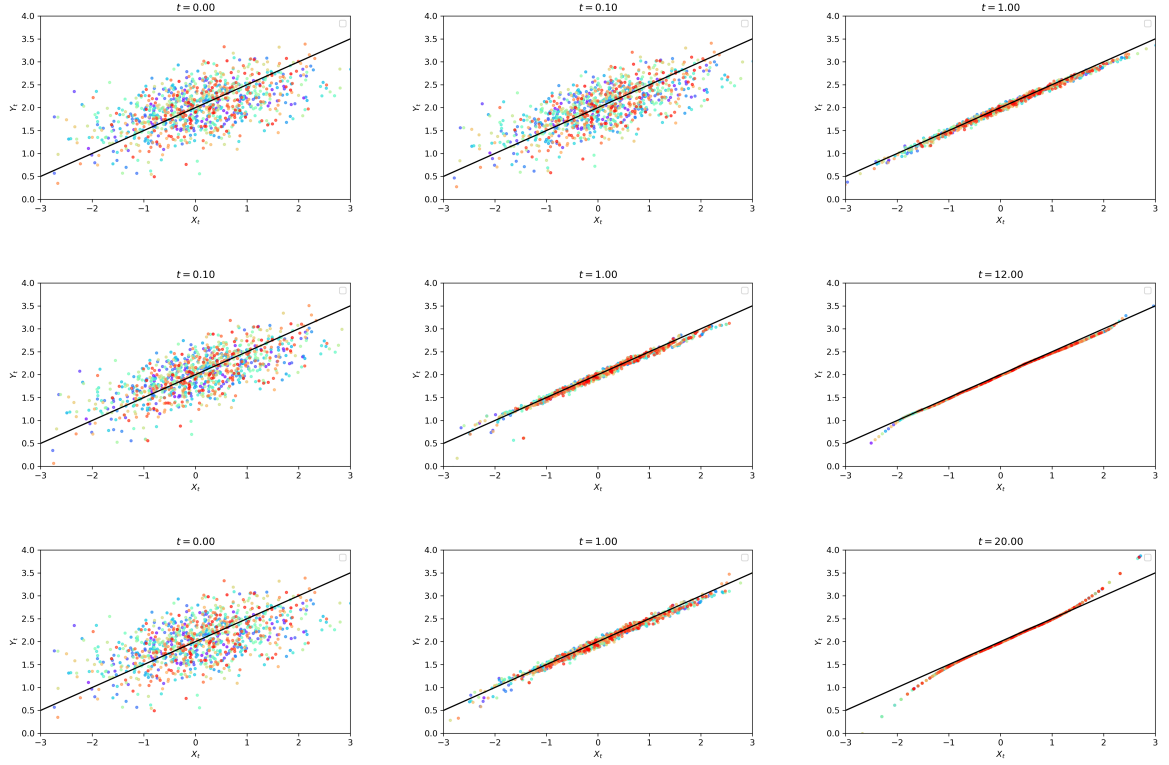


Figure 4: Snapshots of the empirical coupling (X_t, Y_t) (colored dots) versus the true optimal transport map $y = T(x)$ (solid black line) at increasing times. We use $\mu_1 = 0$, $\mu_2 = 2$, $\sigma_1 = 1$, $\sigma_2 = 0.5$ and $\rho = 0.6$. Scheme aparameters: $N = 10^3$, $\varepsilon = 1e - 8$ and $\Delta t = 1.5e - 2$. From top to bottom, the rows correspond to the projection step implemented with (i) *OLS*, (ii) *Nadaraya-Watson* (bandwidth $h = 0.01$), and (iii) *k-NN* ($k = 20$).

for a larger timestep k -NN experience a similar divergence to the NW scheme. By comparing the center plot, it is visible that this error divergence is directly connected to an increase in the estimation error of the conditional expectation. This is predicted by the theory in Sec. 3 where it is stated that the conditional expectation terms play the role of projectors in the valid space of couplings $\Pi(\mu, \nu)$. Finally, the leftmost figure hints at the critical role played by the choice of the entropy regularizer ϵ : the dashed curve (for $\epsilon = 1e-3$) diverges faster than $\epsilon = 1e-8$. This also influences the stochasticity of the results (similarly with Fig. 2). The challenge now is to pinpoint whether this decoupling and the consequent error blow-up stems from one (or more) of the following sources:

1. An inherent limitation of the Nadaraya–Watson or k -NN estimator;
2. Suboptimal choices of the hyperparameters k (for k -NN) and h (for NW);
3. Discretization or sampling error in the projected Langevin scheme, as determined by ϵ , N , and Δt .

We will focus this analysis on the k -NN estimator, since in practice it is the method of choice for comparison with the Sinkhorn algorithm: k -NN admits an $O(N \log N)$ complexity per time step, whereas Nadaraya–Watson incurs $O(N^2)$ cost. It also seems from plots (1) and (2) of Fig. 5 that it might be more robust to the error blow-up.

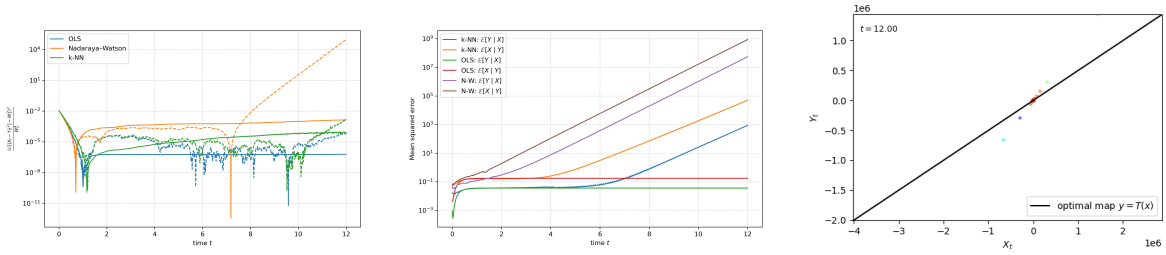


Figure 5: **Left:** Scheme error when using Bures–Wasserstein distance 44 shown for $\epsilon = 1e-8$ (bold line), $\epsilon = 1e-3$ (dashed line). **Center:** the mean-squared error of the conditional-expectation update when the projection is replaced by the exact regression map $m(x) = \mathbb{E}[Y | X = x]$. **Right:** the empirical coupling (X_{20}, Y_{20}) produced by the k -NN projection (with $k = 20$) at final time $t = 20$, plotted against the ideal line $y = T(x)$.

The asymptotically optimal choice of k that minimizes mean-squared error by balancing bias vs. variance (see 5) behaves like:

$$k^* \propto n^{\frac{4}{d+4}}$$

where d demotes the dimension of the samples and this is the value that is used in the subsequent analysis. From the left figure of Figs 6 we can see that at small n (e.g. $n = 10$ or 10^2), the estimator has some order of magnitudes higher relative error even at moderate times, and the blow-up in error occurs earlier. Moreover, the higher the N the better the convergence as it can be seen from the first error minima: this is also true for Δt as it can be seen from fig. 12. As N increases, the curves lie lower but the divergence is not avoided. It might be slightly delayed (see $N = 10^3$ vs. $N = 10^4$) but it is not very clear. From the rightmost figure, we clearly see that smaller ϵ values delay the blow-up showing that lower entropy helps the stability of the scheme. On the other hand, the error curves are virtually equal in their “flat” (stable) portion. Smaller Δt values also result in delayed blow-ups but not as distinctly as ϵ . But they do influence the minima (see fig. 12) by lowering it with smaller Δt s.

Although no choice of n , Δt , or ϵ completely eliminates the eventual blow-up, larger n and smaller Δt reduce the pre-blow-up error, while a smaller ϵ markedly improves stability and delays divergence. This indicates that the blow-up arises probably from an unstable algorithmic implementation rather than from a bad choice of k or h (NW). In principle, one could try a more rigorous hyperparameter search (e.g. cross-validation to pick k), but our current experiments already span many orders of magnitude and show no qualitative improvement. We leave such a CV-based optimization to future work. The stability of the OLS-based projection further reinforces that the problem is not hyperparameter mis-tuning but rather a structural algorithmic weakness of local-averaging in a non-parametric way.

A different possibility that we explored is optimal transport from $\mathcal{N}(0, 1)$ to heavy-tailed exponential distributions $f(y) \propto e^{-y^{2m}}$. The idea is that as there are almost no sample points out in the tails and therefore k -NN regression (or Nadaraya–Watson) has little points over which to average, so its

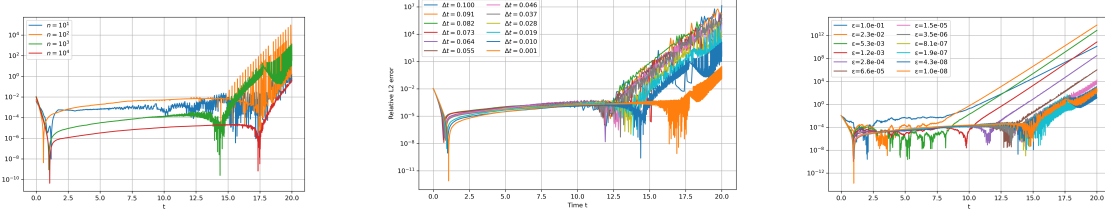


Figure 6: **Left:** Relative L^2 -error over time for different sample sizes $n \in \{10, 10^2, 10^3, 10^4\}$. **Center:** Over time using Δt list = `np.linspace(0.1, 0.001, 12)`. **Right:** Over time using ε list = `np.logspace(-1, -8, 12)`.

conditional expectation estimate near the tails is dominated by variance. But the variance of a k-NN estimate blows up as the target density tends towards zero (see [9], Ch.6):

$$\text{Var}(\hat{m}(x)) \approx \frac{\sigma^2(x)}{k p_X(x)} \quad \text{when } p_X(x) \ll 1 \quad (60)$$

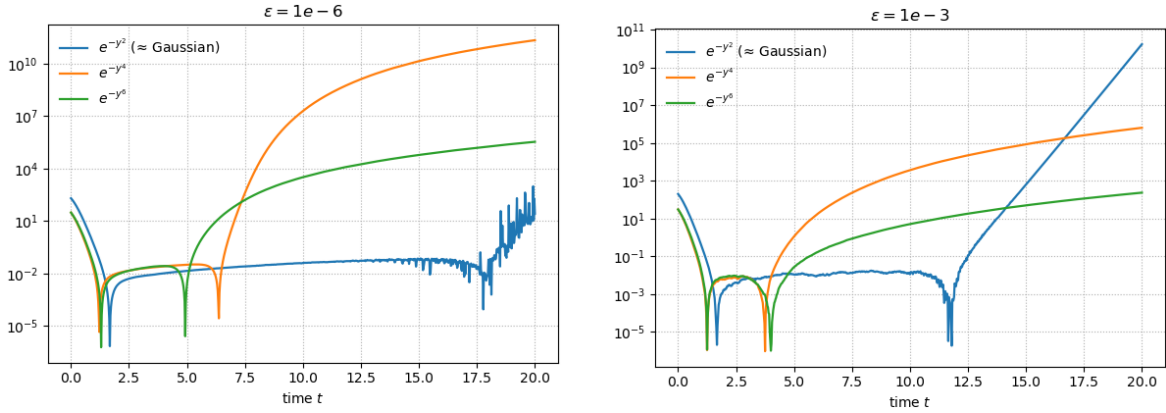


Figure 7: L^2 -error using k -NN from standard Gaussian to target distributions with densities decaying like e^{-y^2} , e^{-y^4} and e^{-y^6} . **Left:** $\varepsilon = 1e - 6$. **Right:** $\varepsilon = 1e - 3$.

From Fig. 7, we can see that the higher the exponential's target degree is, the faster it will experience the error blow-up. But this is not conclusive evidence because we can see that for $\varepsilon = 1e - 3$ the rate at which the blow-up increases for e^{-y^2} is much larger than the other two which is not in accordance with eq. 60.

5.2 Error convergence and Runtimes

We study the asymptotic behavior of the error as well as the runtime depending on the parameters ε and N , for the bivariate Gaussian example given in the previous subsection. In the left Fig.8, ordinary-least-squares (OLS) clearly achieves the lowest relative L^2 error, owing to its exact match with the Gaussian-to-Gaussian Brenier map in this linear setting. The k -nearest-neighbors (k-NN) coupling follows closely: k-NN (and Nadaraya-Watson) make no presupposition on the nature of the regression while on the other hand OLS needs to include the adequate features in its design matrix. This means that the non-parametric methods should be preferred in a more general case (see Fig.11 to see the advantage of the non-parametric methods over OLS). The ε -scaling Sinkhorn solver outperforms Greenkhorn at small ε (e.g. $\varepsilon \lesssim 10^{-1}$ for which Greenkhorn returns `NaN` similarly to fig.1), confirming that warm-start scheduling plus log-stabilized updates are a great improvement to the regular Sinkhorn methods especially for small ε . All five methods show monotonically improving error as $\varepsilon \rightarrow 0$; however, Greenkhorn's error remains pretty high until ε increases similarly to fig.1.

From the right Fig.8, that the the Euler-Maruyama variants (k-NN, OLS, and Nadaraya-Watson) exhibit nearly constant runtime over the entire ε range, because the conditional-expectation regression and time-stepping costs do not depend on the OT regularization (see Sec.3). By contrast, ε -scaling's CPU time grows noticeably as ε decreases, consistent with the theoretical $\mathcal{O}(\log(\varepsilon_0/\varepsilon))$ outer-iteration

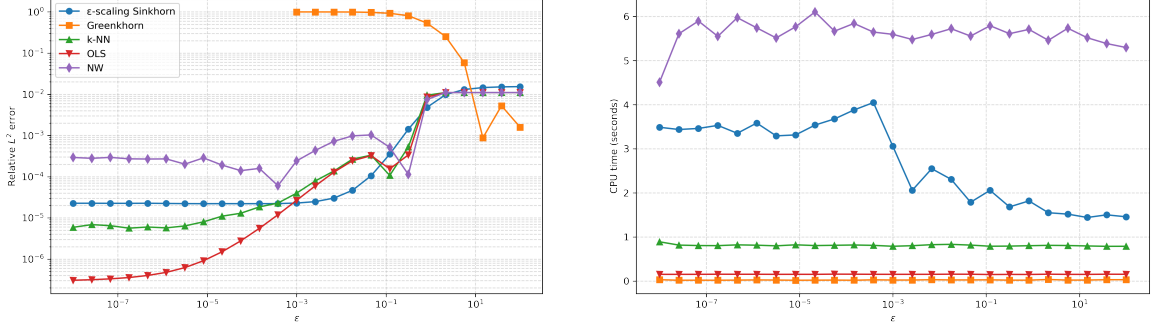


Figure 8: Left: Relative L^2 error versus $\epsilon = \text{np.logspace}(1e1, 1e-8, 25)$ for the five methods (OLS, k-NN, NW, ϵ -scaling, Greenkhorn). **Right:** CPU time in seconds versus ϵ for the same methods.

count (Eq.20). Greenkhorn remains extremely fast as its runtimes are flat and orders of magnitude lower than ϵ -scaling’s for $\epsilon \ll 10^{-1}$. This in agreement with theory that a single row/column update per iteration avoids repeated full-matrix scaling [1].

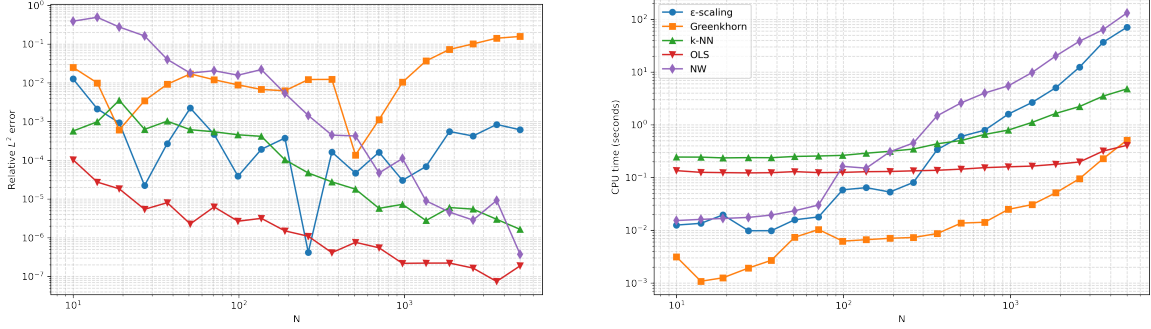


Figure 9: Left: Relative L^2 error versus $N = \text{np.logspace}(1e1, 5e3, 20)$. **Right:** CPU time in seconds versus N .

In the left Fig.9, all three Euler–Maruyama approaches show error decaying sharply with N : both OLS and k-NN drop at a rate of $\mathcal{O}(1/N)$, although OLS is two orders of magnitude lower than k-NN, which in turn is two orders of magnitude lower than Nadaraya–Watson, the latter diminishing at an even faster rate. On the other hand, both ϵ -scaling and Greenkhorn error curves fluctuate a lot, indicating weirdly that their projection-step error does not depend much on the number of samples: adding more points does not significantly reduce the entropic-projection bias at a fixed ϵ . In the right Fig.9, runtime versus N confirms the theoretic complexity rates in Section 4: Nadaraya–Watson scales as $\mathcal{O}(N^2)$, k-NN scales as $\mathcal{O}(N \log N)$, and OLS is essentially $\mathcal{O}(N)$ (with a slight upward bend at large N probablz due to constant-size QR overhead). The ϵ -scaling solver also scales like $\mathcal{O}(N^2)$ respecting the general $\mathcal{O}(N^2)$ established for Sinkhorn–Knopp algorithms (see Sec.2). Greenkhorn remains several orders of magnitude faster than ϵ -scaling and itself grows sub-quadratically (empirically around $\mathcal{O}(N^{1.5})$) because log-stabilized updates converge in fewer iterations. For small N , NW is fastest among the EM methods but has poor error, while for large N , OLS or k-NN are preferred.

5.3 LSI Bound

We conduct our analysis in the case of one-dimensional Gaussian marginals in order to study the behavior of the estimated entropic Wasserstein distance 43

$$\widehat{\mathcal{W}}_t^\epsilon(\mu, \nu) := \mathbb{E}[|X_t - Y_t|^2],$$

and compare it with the theoretical exponential convergence rate established in Theorem 3. We also reference the entropic optimal transport cost $\mathbf{OT}^\epsilon(\mu, \nu)$ defined in Theorem 1.

To match the theoretical envelope to this particular Gaussian setting, we express the predicted upper bound on the coupling error as

$$\widehat{\mathcal{W}}_t^\varepsilon \leq \mathbf{OT}^\varepsilon + \left(\widehat{\mathcal{W}}_0^\varepsilon - \mathbf{OT}^\varepsilon \right) e^{-r(\varepsilon)t} := U_t^\varepsilon$$

where $\widehat{\mathcal{W}}_0^\varepsilon = \mathbb{E}[|X_0 - Y_0|^2]$ is the estimated entropic Wasserstein distance and $r(\varepsilon)$ is the convergence rate derived from Corollary 4. As the marginals (μ, ν) are Gaussian, we have $\alpha_U = \beta_U$ and $\alpha_V = \beta_V$. It can easily be remarked by direct computation $\varepsilon_c = 0$, and as $\varepsilon \rightarrow 0$, the exponent r tends to zero like $\varepsilon(\alpha_U + \alpha_V)/2$ at first order: this is the only category of marginals for which Conforti et al. [6] were able to show exponential convergence for all values of $\varepsilon > 0$.

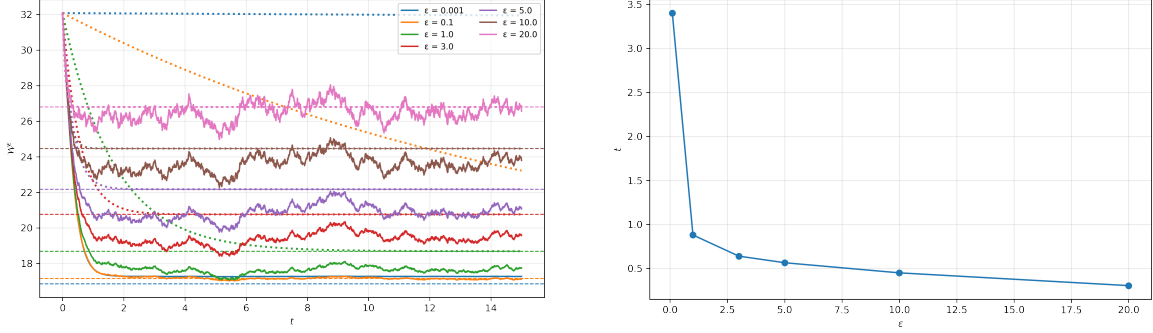


Figure 10: Left: Estimated entropic wasserstein distance $\widehat{\mathcal{W}}_t^\varepsilon(\mu, \nu)$ (bold) against the theoretical exponential bound U_t^ε (dashed) and $\mathbf{OT}^\varepsilon$ (horizontally dashed). Distributions: $\mu = \mathcal{N}(3, 5)$, $\nu = \mathcal{N}(7, 10)$. Parameters: $\Delta t = 5 \times 10^{-3}$, $N_{\text{steps}} = 3 \times 10^3$, $T = N_{\text{steps}} \Delta t = 15$. **Right:** Times $T_{\text{cross}}^\varepsilon$ at which $\widehat{\mathcal{W}}_{T_{\text{cross}}^\varepsilon}^\varepsilon(\mu, \nu) = \mathbf{OT}^\varepsilon$.

From the left figure 10, we observe that the empirical curves remain below their respective envelopes U_t^ε and as ε diminishes they do so at a rate much faster than the one predicted empirically. It is also interesting to note that as ε diminishes the empirical entropic Wasserstein distance falls below $\mathbf{OT}^\varepsilon$: this can be attributed to finite-sample fluctuations. The dependence of the rate on ε is evident: when ε is small the envelope decays slowly, so the crossing times in the right-hand plot are larger, while when ε grows there is rapid decay and earlier plateaux. Finally, the stochastic variability increases with ε because the Euler–Maruyama noise has amplitude $\sqrt{2\varepsilon\Delta t}$ which explains the wider fluctuations of the high- ε trajectories and the smoother behaviour observed for small ε .

6 Conclusion

This report compared entropic-regularised Sinkhorn with a stochastic alternative based on projected Langevin dynamics. After reviewing four modern Sinkhorn variants, we discretised the SDEs given by the projected Langevin dynamics with an explicit Euler–Maruyama scheme and examined three regression algorithms for the conditional-expectation step. Numerical tests in a Gaussian setting showed that ordinary least-squares and k-nearest-neighbour projections achieve smaller transport error than ε -scaling and Greenkhorn in the low- ε regime while preserving linear and near-linear cost respectively per iteration. OLS is fastest and most accurate when the true map is close to linear, whereas kNN offers broader robustness to model uncertainty and still scales as $\mathcal{O}(N \log N)$ per step.

Several questions remain open for future work. A principal issue is the long-time stability of the kNN-driven scheme: simulations reveal that a subset of particle pairs stretches towards infinity after many iterations, and the accuracy of the conditional-expectation estimate degrades in parallel, leading to runaway coupling error.

Understanding whether this phenomenon stems from estimator variance, from the discretisation, or from an intrinsic instability of the projected flow is essential because kNN currently offers the best balance between accuracy and robustness. Another challenge concerns high-dimensional data: the kd-tree acceleration underpinning the kNN search deteriorates as dimension grows, a manifestation of the curse of dimensionality that both slows queries and inflates statistical error rates.

Extending the analysis to real-world tasks where marginal laws are unknown is equally important, since continuity assumptions required for theoretical guarantees must then be verified or approximated

empirically. Finally, Conforti et al. [6] identified marginals that admit a critical threshold ε_c , below which the theoretical convergence rate is unknown. Characterising the dynamics of such under-critical systems for $\varepsilon < \varepsilon_c$ remains an open mathematical problem.

Appendix A Algorithms

Algorithm 7: NN-KD-TREE-QUERY [19]

Input: Query point $X \in \mathbb{R}^d$, k-d tree T , threshold ε , number of neighbors k
Output: List L of k nearest neighbors to X within distance ε

```

1  $L \leftarrow$  empty list
2 return NN-KD-TREE-QUERY( $X, T.root, \varepsilon, k, L$ )

3 Recursive Procedure NN-KD-TREE-QUERY( $X, T, \varepsilon, k, L$ ):
4 if  $T = \emptyset$  then
5   | return  $L$ 
6 end
7 Extract point  $X_j$  from  $T.root$  and store it in  $L$  if  $\|X_j - X\| \leq \varepsilon$ 
8 if  $|L| > k$  then
9   | Delete from  $L$  the point farthest from  $X$ 
10 end
11 if  $T$  is a leaf node then
12   | return  $L$ 
13 end
14 if  $T.root.leftSubtree$  contains  $X$  then
15   |  $T_1 \leftarrow T.root.leftSubtree, T_2 \leftarrow T.root.rightSubtree$ 
16 else
17   |  $T_1 \leftarrow T.root.rightSubtree, T_2 \leftarrow T.root.leftSubtree$ 
18 end
19 NN-KD-TREE-QUERY( $X, T_1, \varepsilon, k, L$ )
20 if  $|L| < k$  and  $\|X - T.root.hyperplane\| < \varepsilon$  then
21   | NN-KD-TREE-QUERY( $X, T_2, \varepsilon, k, L$ )
22 end
23 else if  $\|L.farthest - X\| > \|X - T.root.hyperplane\| \leq \varepsilon$  then
24   | NN-KD-TREE-QUERY( $X, T_2, \varepsilon, k, L$ )
25 end
26 return  $L$ 

```

Appendix B Figures

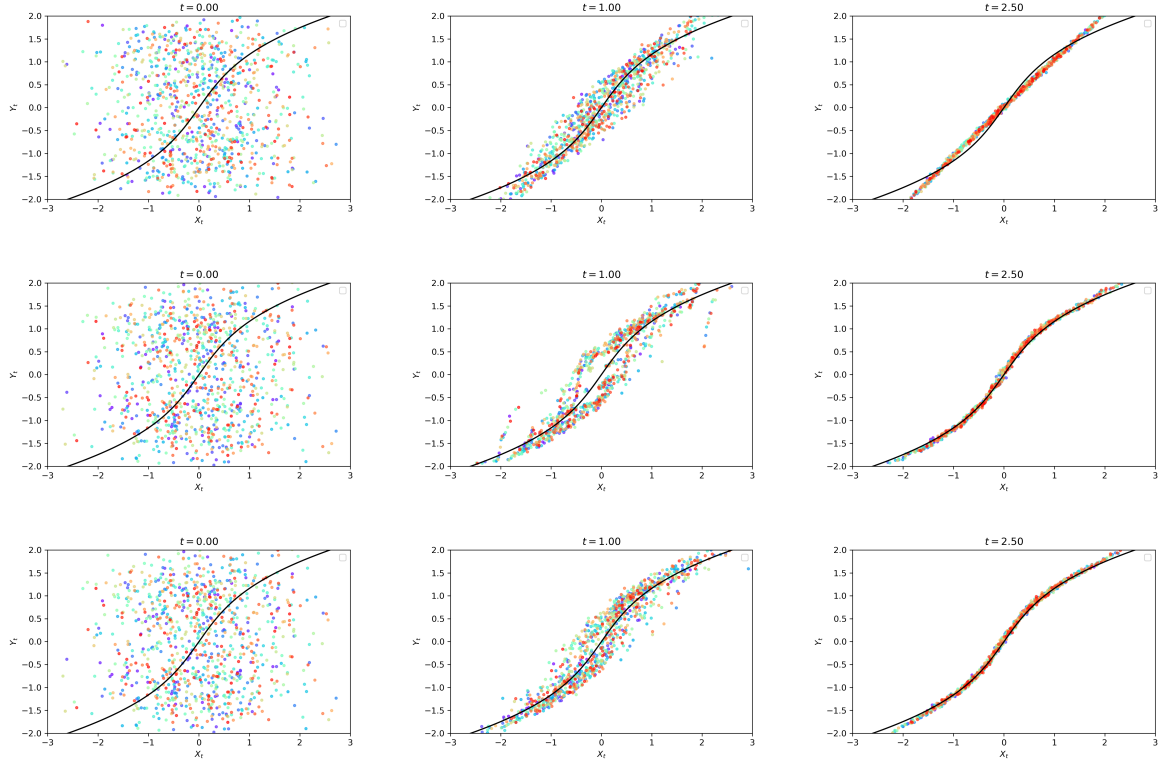


Figure 11: Snapshots of the coupling scatter (X_t, Y_t) against the optimal map $y = T(x)$ at times $t = 0.00, 1.00, 2.50$, for marginals $\mu = \mathcal{N}(0, 1)$ and $\nu = \exp -(y^4/4 + y^2/2)$. From top to bottom: results using the *OLS* conditional-expectation estimator; the *Nadaraya-Watson* estimator (bandwidth $h = 0.01$); and the *k-NN* estimator ($k = 20$ neighbors).

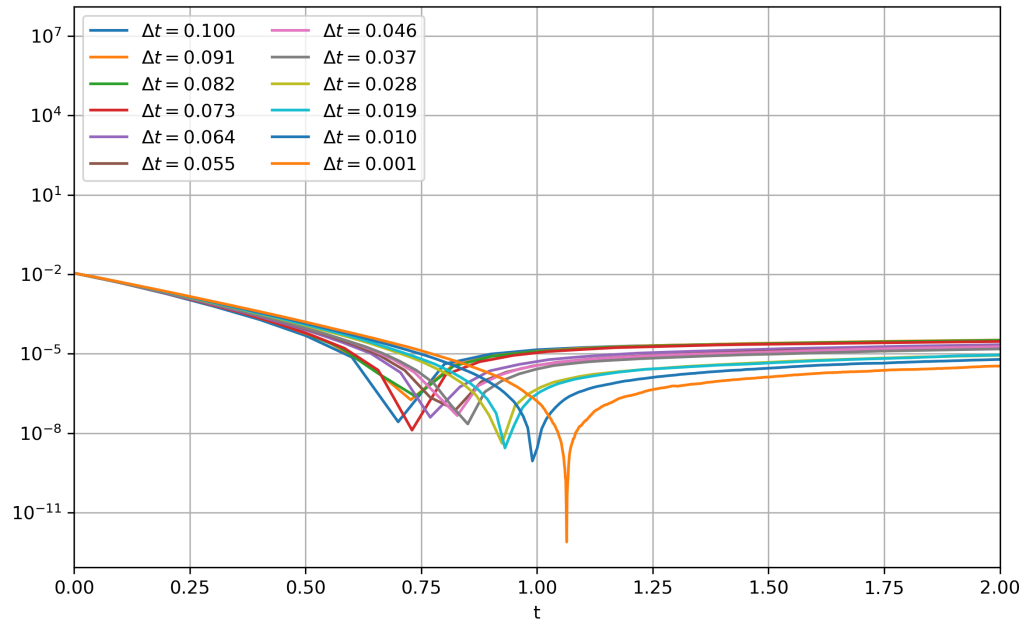


Figure 12: Zoom of center figure 6.

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